DIET AND EXERCISE (HHSCI)

OVERVIEW

Diet and Exercise, B.S./M.S.

Administered by the Department of Food Science and Human Nutrition and Department of Kinesiology and Health.

This is an accelerated program with concurrent enrollment in the undergraduate and graduate degree programs. Courses included have been approved as meeting the academic requirements of the Didactic Program in Dietetics (DPD) in preparation for admission to accredited dietetics internship programs; the DPD is accredited by the Accreditation Council for Education in Nutrition and Dietetics, the accrediting agency of the Academy of Nutrition and Dietetics. There is a $30 fee for the verification statement of completion of the accredited dietetics program. Additionally, courses are included to meet the American College of Sports Medicine (ACSM) requirements for certification at the level of Certified Exercise Physiologist.

The Diet and Exercise program of study is not offered as an additional major, it is only offered as a full degree program. Students pursuing the Diet and Exercise degree program are allowed to pursue additional majors in other eligible programs of study if desired.

Student Learning Outcomes

Upon graduation, students should be able to:

- Communicate effectively in their field of study using written, oral, visual and/or electronic forms.
- Demonstrate proficiency in ethical data collection and interpretation, literature review and citation, critical thinking and problem solving.
- Participate effectively in a group or team.
- Integrate creativity, innovation, or entrepreneurship in ways that produce value.
- Explain how human activities impact the natural environment and how societies are affected.
- Meet program specific learning outcomes for the Diet & Exercise major.

Degree Requirements

Total Degree Requirements: 120 cr. for bachelor's degree and 36-40 cr. for master's degree

International Perspectives: 3 cr.

U.S. Cultures and COmmunities: 3 cr.

Students must fulfill International Perspectives and U.S. Cultures and Communities requirements by selecting coursework from approved lists. These courses may also be used to fulfill other area requirements.

Communications and Library: 10 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
LIB 1600	Introduction to College Level Research	1
SPCM 2120	Fundamentals of Public Speaking	3
Total Credits		**10**

Social Sciences: 6 cr.

PSYCH 1010	Introduction to Psychology	3
KIN 3660	Exercise Psychology	3
Total Credits		**6**

Mathematical Sciences: 6-8 cr.

Select at least 3 credits from:		3-4
MATH 1400	College Algebra	
MATH 1430	Preparation for Calculus	
MATH 1600	Survey of Calculus	
MATH 1650	Calculus I	
Select at least 3 credits from:		3-4
STAT 1010	Principles of Statistics	
STAT 1040	Introduction to Statistics	
STAT 2026	Introduction to Business Statistics	
Total Credits		**6-8**

Physical Sciences: 13 cr.

Select from:		5
CHEM 1630 & 1630L	College Chemistry and Laboratory in College Chemistry	
or CHEM 177 & 1770L	General Chemistry I and Laboratory in General Chemistry I	
CHEM 2310	Elementary Organic Chemistry	3
CHEM 2310L	Laboratory in Elementary Organic Chemistry	1
PHYS 1150	Physics for the Life Sciences	4
or PHYS 1310	General Physics I	
Total Credits		**13**

Biological Sciences: 13 cr.

BBMB 3010	Survey of Biochemistry	3
BIOL 2550	Fundamentals of Human Anatomy	3
BIOL 2550L	Fundamentals of Human Anatomy Laboratory	1
BIOL 2560	Fundamentals of Human Physiology	3
BIOL 2560L	Fundamentals of Human Physiology Laboratory	1
MICRO 2010	Introduction to Microbiology	2
Total Credits		**13**

Diet and Exercise undergraduate courses to be completed or in progress when applying for admission to the program: 21-23 cr.

Select from:		1-2
FSHN 1100	Professional and Educational Preparation	
or KIN 2520	Introduction to the Discipline of Kinesiology and Orientation and Learning Community in	
& KIN 2530	Kinesiology and Health	
FSHN 1670	Introductory Human Nutrition and Health	3
FSHN 2140	Scientific Study of Food	3
FSHN 2150	Advanced Food Preparation Laboratory	1-2
or FSHN 1150	Food Preparation Laboratory	
FSHN 2650	Metabolism, Nutrition and Health	3
FSHN 3400	Foundations of Dietetic Practice	2
FSHN 3600	Advanced Nutrition and the Regulation of Metabolism in Health and Disease	3
HS 1100	Personal and Consumer Health	3
KIN 2580	Principles of Physical Fitness and Conditioning	2
Total Credits		**21-23**

Apply to the B.S./M.S. program by stated deadlines.

Humanities and Ethics: 6-9 cr.

Select 6 credits from approved Humanities list	6
Select 3 credits from approved Ethics list	3
Note: If ethics course is on the humanities list, it can meet both requirements.	

Diet and Exercise remaining undergraduate courses to complete the bachelor's degree requirements: 42 cr.

HS 3800	Worksite Health Promotion	3
or HS 3500	Human Diseases	
ATR 2200	Basic Athletic Training	2
or HS 1050	First Aid and Emergency Care	
KIN 2590	Leadership Techniques for Fitness Programs	3
KIN 3450	Management of Health-Fitness Programs and Facilities	3
or FSHN 3920	Food and Nutrition Services Management	
KIN 3580	Exercise Physiology	3
KIN 3590	Exercise Physiology Lab	1
Select two:		6
KIN 3550	Biomechanics	
KIN 3600	Sociology of Physical Activity and Health	
KIN 3720	Motor Control and Learning Across the Lifespan	
KIN 4580	Principles of Fitness Assessment and Exercise Prescription	
KIN 4800	Functional Anatomy	
KIN 4620	Medical Aspects of Exercise	3
FSHN 3610	Nutrition and Health Assessment	2
FSHN 3670	Medical Terminology for Health Professionals	1
FSHN 3800	Food Production Management	3
FSHN 3800L	Food Production Management Experience	3
FSHN 4300	U.S. Health Systems and Policy	2
FSHN 4660	Nutrition Counseling and Education Methods	3
HSPM 1330	Food Safety Certification	1
NUTRS 5630	Community Nutrition and Health *	3
Total Credits		**42**

Diet and Exercise graduate courses to complete the master's degree requirements: 36-40 cr.

FSHN 5900C	Special Topics: Teaching	1
FSHN 6810	Seminar **	1
FSHN 6820	Research Presentation and Discussion (repeatable, required each semester as graduate student) **	1
NUTRS 5100	Advanced Nutrition and Metabolism in Health and Disease: Macronutrients	2
NUTRS 5110	Advanced Nutrition and Metabolism in Health and Disease: Micronutrients	2
NUTRS 5610	Medical Nutrition and Disease I	4
NUTRS 5630	Community Nutrition and Health *	3
NUTRS 5640	Medical Nutrition and Disease II	4
KIN 5010	Research Methods in Physical Activity	3
KIN 5050	Research Laboratory Techniques in Exercise Physiology	3
KIN 5500	Advanced Bioenergetics of Exercise	3
or KIN 5510	Advanced Cardiorespiratory Exercise Physiology	
STAT 5101	Statistical Methods for Data Analysis	4
FSHN select 3 additional credits, KIN select 6 additional credits from:		3-6
KIN 5670	Exercise and Health: Behavior Change	
KIN 5500	Advanced Bioenergetics of Exercise	
KIN 5510	Advanced Cardiorespiratory Exercise Physiology	
KIN 5700	Physical Activity Assessment for Health Related Research	
KIN 5730X	Impact of Physical Activity on Healthy Aging	
Select 2-3 credits for creative component or 6 credits for thesis research:		2-6
FSHN 5990	Creative Component	
or KIN 5990	Creative Component	
KIN 6990	Research	
or NUTRS 6990	Research in Nutritional Sciences	

* Course counts toward both bachelor's and master's degrees.

** Requirement for students in the FSHN Department.

Go to FSHN courses. (https://catalog.iastate.edu/azcourses/fs_hn/)

Go to KIN courses. (https://catalog.iastate.edu/azcourses/kin/)

Diet and Exercise, B.S./M.S.

First Year

Fall	Credits	Spring	Credits
FSHN 1100 or KIN 2520 ***and*** KIN 2530	1-2	FSHN 1670	3
CHEM 1630 or 1770	4	ENGL 2500	3
CHEM 1630L or 1770L	1	HS 1100	3
ENGL 1500	3	KIN 2580	2
LIB 1600	1	STAT 1010, 1040, or 2026	3-4
MATH 1400, 1430, 1600, or 1650	3-4		
PSYCH 1010	3		
	16-18		**14-15**

Second Year

Fall	Credits	Spring	Credits	Summer	Credits
BIOL 2550	3	FSHN 2650	3	SPCM 2120	3
BIOL 2550L	1	BIOL 2560	3		
KIN 3660	3	BIOL 2560L	1		
CHEM 2310	3	BBMB 3010	3		
CHEM 2310L	1	FSHN 2140	3		
KIN 2590	3	FSHN 1150 or 2150	1-2		
HSPM 1330	1	MICRO 2010	2		
	15		**16-17**		**3**

Third Year

Fall	Credits	Spring	Credits	Summer	Credits
FSHN 3400	2	FSHN 3610	2	KIN 5990, FSHN 5990, KIN 6990, or NUTRS 6990	1-3
FSHN 3600	3	FSHN 3670	1	STAT 5101	4
KIN 3580	3	FSHN 3800	3		
KIN 3590	1	FSHN 3800L	3		
ATR 2200 or HS 1050	2	Select from:	6-7		
PHYS 1150 or 1310	4	KIN 3550, 3660, 3720, 4580, or 4800			
Apply for admission to the BS/MS program by Oct. 1st.					
Acceptance into the program required before spring of the third year.					
	15		**15-16**		**5-7**

Fourth Year

Fall	Credits	Spring	Credits	Summer	Credits
KIN 5050	3	KIN 5010	3	FSHN 5990, KIN 5990, KIN 6990, or NUTRS 6990	1-3
NUTRS 5610	4	KIN 5510	3		
NUTRS 5630*	3	NUTRS 5640	4		
FSHN 6820	1	FSHN 6820	1		
KIN 5500, 5510, 5670, 5700, or 5730X	3	KIN 4620	3		
	14		**14**		**1-3**

Fifth Year

Fall	Credits	Spring	Credits
NUTRS 5100	2	NUTRS 5110	2
FSHN 4300	2	FSHN 6810	1
FSHN 5900C	1	FSHN 6820	1

FSHN 6820	1	KIN 3450 or FSHN 3920	3
KIN 5670, 5500, 5510, 5700, or 5730X	3	KIN 6990, NUTRS 6990, KIN 5990, or FSHN 5990	2
HS 3500 or 3800	3	FSHN 4660	3
Humanities/ Ethics course	3	Humanities/ International Perspectives	3
	15		**15**

Planned course offerings may change, and students need to check the online Schedule of Classes each term to confirm course offerings: http://classes.iastate.edu/. This sequence is only an example.

* NUTRS 5630 counts toward both bachelor's and master's degrees. Students must have at least 120 credits for the bachelor's degree and at least 30 credits for the master's degree.